### 8.2.2 Statistical Probability (Relative Frequency/a posteriori Approach)

If the random experiment is repeated n times under identical conditions and the event A occurred in n(A) times, then the probability for the occurrence of the event A can be defined (Von Mises) as

$$P(A) = \lim_{n \to \infty} \frac{n(A)}{n}.$$

Since computation of probability under this approach is based on the empirical evidences for the occurrence of the event, it is also knows as relative frequency or a posteriori probability.

**NOTE**

It should be noted that repeating some experiments is not always possible. In such cases, relative frequency approach cannot be applied. Classical approach to compute probability requires that the sample space should be a finite set. It is seldom possible. Hence, we present in the next section axiomatic approach to probability which overcomes the limitations of both mathematical probability and statistical probability.

## 8.3 Axioms of Probability



**(A.N. Kolmogorov)**

Andrey Nikolaevich Kolmogorov (1903–1987) was a 20th-century Soviet mathematician who made significant contributions to the mathematics of probability theory, topology, intuitionistic logic, turbulence, classical ... Wikipedia

A.N. Kolmogorov proposed the axiomatic approach to probability in 1933. An axiom is a simple, indisputable statement, which is proposed without proof. New results can be found using axioms, which later become as theorems.

### 8.3.1 Axiomatic approach to probability

Let $S$ be the sample space of a random experiment. If a number $P(A)$ assigned to each event $A \in S$ satisfies the following axioms, then $P(A)$ is called the probability of $A$.

Axiom-1 : $P(A) \geq 0$

Axiom-2 : $P(S) = 1$

Axiom-3 : If $\{A_1, A_2, \ldots\}$ is a sequence of mutually exclusive events i.e., $A_i \cap A_j = \phi$